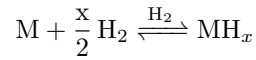


2. Physics

In this section we will be looking at some of the physics behind the equations we will be using in our models. We start by looking at the absorption and desorption of Metal Hydride Materials in Section 2.1, and explain how we get to the absorption equations in our models. Then we will be looking at the flow of our Hydrogen gas in Section 2.2, as these will be the most important equations for our models.

2.1. Absorption and Desorption in Metal Hydride Materials

Metal Hydrides have been researched extensively. They are materials that can absorb Hydrogen gas, and store it in a solid form. This is done by the metal hydride absorbing the hydrogen gas, and forming a chemical bond with it. This is a reversible process, and the hydrogen can be released from the metal hydride by heating it up. This makes metal hydrides a good candidate for hydrogen storage, as they can store large amounts of hydrogen in a small volume. The chemical reaction for the absorption and desorption of hydrogen in metal hydrides is surprisingly simple, and can be given by the following chemical reaction[4]:



Where M is the metal, H_2 is the hydrogen gas, and MH_x is the metal with Hydrogen attached to it. x is the number of hydrogen atoms that can be absorbed by the metal. The absorption and desorption of hydrogen in metal hydrides is a reversible, but has a hysteresis effect, meaning that the absorption and desorption curves are not the same. A schematic of the absorption and desorption curves can be seen in Figure 1. From now on if we talk about absorption, we will be looking at the upper part of the curve, and if we talk about desorption, we will be looking at the lower part of the curve. For these parts, the mass absorption, and desorption rates are given by[6]:

$$\dot{m}(\rho, \rho_s) = C_a \exp\left(-\frac{E_a}{RT}\right) \log\left(\frac{p_p}{p_{eq,a}}\right) (\rho_{sat} - \rho_s(t)), \quad (1)$$

$$\dot{m}(\rho, \rho_s) = C_d \exp\left(-\frac{E_d}{RT}\right) \left(1 - \frac{p_p}{p_{eq,d}}\right) (\rho_s(t) - \rho_e), \quad (2)$$

where C_a and C_d are the absorption and desorption rate constants in $[s^{-1}]$, E_a and E_d are the activation energies for absorption and desorption in $[J/mol]$, R is the universal gas constant in $[J/(mol K)]$, T is the temperature, p_p is the pressure of the hydrogen gas, $p_{eq,a}$ and $p_{eq,d}$ are the equilibrium pressures for absorption and desorption, ρ_{sat} is the saturation density of the metal hydride, $\rho_s(t)$ is the density of the metal hydride at time t , and ρ_e is the initial density of an empty battery.

In both papers they also calculate the equilibrium pressures for absorption and desorption using the Van't Hoff equation, which is given by:

$$\log\left(\frac{p_{eq}}{P_{ref}}\right) = A - \frac{B}{T} \quad (3)$$

Where P_{ref} is the reference pressure set to $1MPa$, A and B are constants that depend on the metal hydride, set to 10.7 and 3704.6K for absorption, and 10.57 and 3704.6K for desorption.

In all our models we will assume constant temperature, and we will be looking at the absorption and desorption of hydrogen gas in metal hydrides. This leads us to simplify Equations 1 and 2 to the following equations:

$$\dot{m}(\rho_g, \rho_s) = A_a \log\left(\frac{p}{p_{eq,a}}\right) (\rho_{sat} - \rho_s(t)) \quad (4)$$

$$\dot{m}(\rho_g, \rho_s) = A_d \left(1 - \frac{p}{p_{eq,d}}\right) (\rho_s(t) - \rho_o), \quad (5)$$

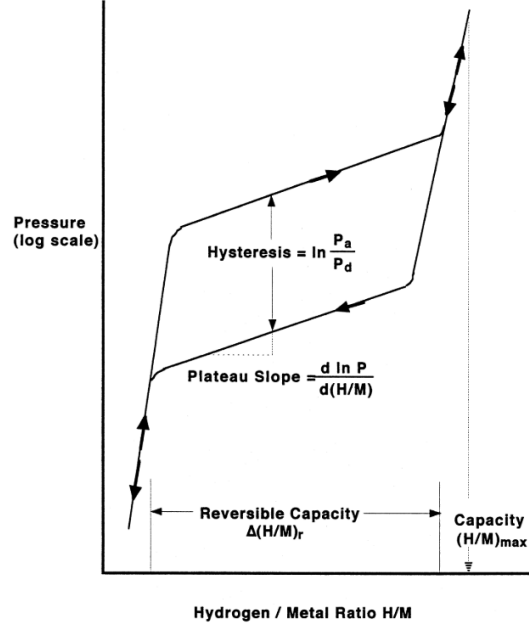


Figure 1: Schematic of the absorption and desorption curves of metal hydrides. The x-axis is the ratio between Hydrogen and Metal, and the capacity can therefore be interpreted as a measure for x in MH_x . The absorption curve is the left and upper lines, the desorption is given in the right and lower lines. The hysteresis effect can be seen in the difference between the two curves. Schematic from Sandrock et al.[4]

for absorption and desorption respectively. Here A_a and A_d are constants that depend on the temperature, and the properties of the metal hydride, and are given by:

$$A_a = C_a \exp\left(-\frac{E_a}{RT}\right) \quad (6)$$

$$A_d = C_d \exp\left(-\frac{E_d}{RT}\right). \quad (7)$$

Both in $[s^{-1}]$.

We will be looking at absorption and desorption in our models, for simplicity, we assume that charging (absorption) of the battery always starts from an empty battery, and discharging (desorption) of the battery always starts from a full battery. This gives the following initial conditions for the absorption and desorption models:

$$\rho_s(0) = \rho_e, \quad (8)$$

$$\rho_s(0) = \rho_{sat}, \quad (9)$$

For pressure we assume absorption starts at the equilibrium pressure for absorption (at that temperature), and desorption starts at the equilibrium pressure for desorption (at that temperature). Both calculated using Equation 3. Leading to initial conditions for the pressure:

$$p_p(0) = p_{eq,a}, \quad (10)$$

$$p_p(0) = p_{eq,d}, \quad (11)$$

2.2. Flow

The governing equations consist of the conservation equations (mass, momentum and energy) together with constitutive equations describing the fluid properties. Assuming a Newtonian fluid, the constitutive relation for the stress tensor leads to the Navier-Stokes equations. The conservation equations are derived from the fundamental laws of physics, while the constitutive equations describe the material properties of the fluid. We will not be looking at the energy equation, as we assume constant temperature in our models.

The conservation of mass equation is given by (in its continuous form):

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = \dot{m} \quad (12)$$

Where ρ is the density of the fluid, $\mathbf{u}$ is the velocity of the fluid, and $\dot{m}$ is the mass source term. The mass source term is given by the absorption and desorption rates we derived in Section 2.1. The first term represents the local rate of change of mass, while the second term represents the net mass flux leaving the control volume. Together, they equal the volumetric mass source term due to absorption or desorption.

For Navier-Stokes, I like a little derivation starting from Newton's second law. Lets look at a small volume of fluid, now we get:

$$\begin{aligned} \mathbf{F} &= m \mathbf{a} \\ &= m \frac{D\mathbf{u}}{Dt} \\ &= m (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) \\ \implies \frac{\mathbf{F}}{V} &= \rho (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) \end{aligned}$$

where $\mathbf{F}$ is the force, m is the mass, $\mathbf{a}$ is the acceleration, V is the volume, and ρ is the density. Common forces acting on the fluid are pressure, and the viscous forces. Also, for a porous medium such as a metal hydride bed, the fluid experiences an additional resistance due to friction with the solid matrix. Darcy observed experimentally that this resistance is proportional to the fluid velocity, leading to Darcy's law[7]. These are given by:

$$\begin{aligned} \frac{\mathbf{F}_p}{V} &= -\nabla p \\ \frac{\mathbf{F}_v}{V} &= \mu \Delta \mathbf{u} \\ \frac{\mathbf{F}_e}{V} &= -\frac{\mu}{K} \mathbf{u} \end{aligned}$$

Where p is the pressure, μ is the dynamic viscosity in $[\text{kg}/(\text{m s})]$, and K is the permeability of the porous medium in $[\text{m}^2]$. For the viscosity, we did assume a Newtonian fluid. Combining above equations, we get to our version of the Navier-Stokes equation:

$$\rho (\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = -\nabla p + \mu \Delta \mathbf{u} - \frac{\mu}{K} \mathbf{u} \quad (13)$$

To go from Navier-Stokes to Stokes, we assume that the flow is slow, and the inertial forces are negligible compared to the viscous forces. This is known as the low Reynolds number assumption. The Reynolds number is a dimensionless quantity that describes the ratio of inertial forces to viscous forces in a fluid flow. It is given by:

$$Re = \frac{\rho U L}{\mu} \quad (14)$$

Where U is the characteristic velocity, and L is the characteristic length. For low Reynolds number flows, the inertial forces are negligible compared to the viscous forces, and the Navier-Stokes equation reduces to the Stokes equation:

$$\rho_g \partial_t \mathbf{u} = -\nabla p + \mu \Delta \mathbf{u} \quad (15)$$

Pressure

We will be looking at a pure hydrogen gas, and we will be looking at the density of the hydrogen gas ρ_g , which will also (as described in the previous section) be related to the pressure of the hydrogen gas p through the ideal gas law. The ideal gas law is given by:

$$p = RT\rho_g \quad (16)$$

where R is the specific gas constant for hydrogen, equal to $R = 8.314 \text{ J}/(\text{mole K})$. Instead of prescribing the static pressure at the inlet, a total pressure boundary condition will later also be adopted[8],

$$p_{\text{tot}} = p_g + \frac{1}{2}\rho_g|\mathbf{v}|^2, \quad (17)$$

where p_g is the static pressure and $\frac{1}{2}\rho_g|\mathbf{v}|^2$ is the dynamic pressure. In practice, the pressure supplied by a gas reservoir or pressure regulator corresponds to the total pressure of the flowing gas rather than its static pressure. Prescribing the total pressure therefore allows the static pressure at the inlet to decrease naturally when the gas velocity increases, while recovering the prescribed value when the flow is at rest. This provides a more realistic description of hydrogen entering the porous hydride bed, particularly during the initial stage of absorption when the inflow velocity is highest. Since the absorption kinetics depend on the static hydrogen pressure through the driving force $\ln(p_g/p_{eq})$, using a total pressure boundary condition results in a more physically consistent coupling between the inlet flow and the reaction rate.

2.3. Model Assumptions

In this section we will be looking at the assumptions we make in our models. The models are supposed to increase in complexity, and accuracy. We start with a 0D model, where we do not look at the flow of the hydrogen gas. In our next iteration, we will be looking at the same mass absorption and desorption equations, but we will be looking at the flow of the hydrogen gas through a 1D battery. Here we will be assuming constant flow and density for the movement of the gas, and our only focus will be on the diffusion, convection, and absorption/desorption of the hydrogen gas. We replateate these results with the same equations in a 2D battery in our third model. In our fourth model, we will be looking at the flow of the hydrogen gas through a 1D battery, but now we will be looking at the full Navier-Stokes equations, and we will be looking at the pressure of the hydrogen gas as well. We will also be looking at the absorption and desorption of the hydrogen gas in the metal hydride. In our fifth model, we will be looking at the same equations as in our fourth model, but we will be looking at a 2D battery.

We want to model everything until the battery is fully charged or fully discharged. Because there is exponential growth we will stop when we are within xxx% of a full/empty battery, which we will refer to as full/empty.

The governing equations described in the above sections form a coupled system describing the evolution of the hydrogen gas and the metal hydride. Depending on the model, the unknowns are the hydrogen gas density ρ_g , the hydrogen velocity $\mathbf{u}$, and the density of absorbed hydrogen in the metal hydride ρ_s . The governing equations consist of one mass conservation equation for ρ_g , one momentum equation for each velocity component, and one local evolution equation for ρ_s . Together with the constitutive relations for the mass transfer rate and the ideal gas law, the number of equations equals the number of unknowns, resulting in a closed system.

Model 1: perfectly mixed battery

In our first model, we aim to compute the density of hydrogen gas ρ_g in the battery, and the saturation of the metal hydride ρ_s . We do not yet aim to compute velocity of the gas and it is assumed that the flow is perfectly mixed. This gives us:

$$\frac{D\rho_g}{Dt} = -\dot{m}(\rho_g, \rho_s) \quad (18a)$$

$$\frac{D\rho_s}{Dt} = \dot{m}(\rho_g, \rho_s) \quad (18b)$$

With m given by Equations 4 and 5, for absorption and desorption respectively. We also apply the ideal gas law, Equation 16, to write the governing equations in terms of ρ_g . Initial conditions are given by:

$$\rho_g(0) = \rho_{g,0}, \quad (19a)$$

$$\rho_s(0) = \rho_{s,0}. \quad (19b)$$

where $\rho_{s,0}$ is the initial density of the metal hydride in Equations 8 and 9, and $\rho_{g,0}$ is the equilibrium density corresponding to the equilibrium pressure for absorption or desorption in Equations 10 and 11.

Model 2: 1D battery, constant flow and pressure

For our first spacially coupled models, we aim to compute the density of hydrogen gas ρ_g in the battery, and the saturation of the metal hydride ρ_s . We do not yet aim to compute velocity of the gas and it is assumed that the flow is constant through the battery. The pressure is assumed to vary a little bit, leaving us to be able to neglect it in the momentum equations, but not in the mass transfer functions (since there a small variation is the difference between no absorption and absorption). The model is 1D and we will be looking at the domain $\Omega = (0, L)$, where L is the length of the battery. The left boundary will be the inflow of hydrogen gas, and the right boundary will be the outflow of hydrogen gas. The model is shown in Figure 2.

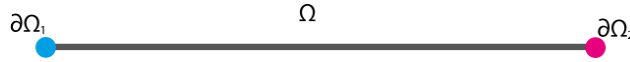


Figure 2: The domain of the 1D model. The domain is Ω . The boundary $\partial\Omega_1$ is for the inlet of the battery.

The above assumptions simplify our models a lot, and allow us to focus on the diffusion and convection of the hydrogen gas:

$$\partial_t \rho_g = D \partial_x^2 \rho_g - v \partial_x \rho_g - \dot{m}(\rho_g, \rho_s) \quad (20a)$$

$$\partial_t \rho_s = \dot{m}(\rho_g, \rho_s) \quad (20b)$$

where we assume constant flow v through the battery, and D is the diffusion coefficient of the hydrogen gas. The initial conditions are given by:

$$\rho_g(x, 0) = \rho_{g,0}, \quad (21a)$$

$$\rho_s(x, 0) = \rho_{s,0}. \quad (21b)$$

which can be found in the same methods as in Model 1. The boundary conditions are given by:

$$\rho_g(\partial\Omega_1, t) = \rho_{g,in}, \quad (22a)$$

$$\partial_x \rho_g(\partial\Omega_2, t) = 0. \quad (22b)$$

where $\rho_{g,in}$ is the density of the hydrogen gas at the inlet of the battery, and L is the length of the battery. The Dirichlet boundary condition represents the fact that we have a constant density of hydrogen gas into the battery, and the Neumann boundary condition represents the fact that we have no flux of hydrogen gas out of the battery.

Model 3: 2D battery, constant flow and pressure

This part is meant to combine absorption, diffusion, and convection shown in the previous model to a 2D model. The shape of the model is given in Figure 3.

We use the same physics and assumptions as in Model 2, but rewrite the governing equations in 2D. Equation 20 becomes:

$$\partial_t \rho_g = D \Delta \rho_g - \mathbf{v} \cdot (\nabla \rho_g) - \dot{m}(\rho_g, \rho_s) \quad (23a)$$

$$\partial_t \rho_s = \dot{m}(\rho_g, \rho_s) \quad (23b)$$



Figure 3: The domain of the 2D model. The domain is Ω . The boundary $\partial\Omega_1$ is for the inlet of the battery. The boundary $\partial\Omega_2$ is for the insulating boundary.

where now $\mathbf{v}$ is a vector with the x-component and y-component of the velocity, and Δ is the Laplace operator. The initial conditions are given by:

$$\rho_g(\mathbf{x}, 0) = \rho_{g,0}, \quad (24a)$$

$$\rho_s(\mathbf{x}, 0) = \rho_{s,0}. \quad (24b)$$

Model 4: 1D Navier Stokes

Our next model is to look at more realistic behaviour for the flow of hydrogen gas. We start by a one dimensional implementation of the Navier Stokes equation, with a mass conservation equation where mass transfer is included. The model is 1D and we will be looking at the same domain as model 2. We model the flow of hydrogen v , the density of hydrogen gas ρ_g , and the saturation of the metal hydride ρ_s . The governing equations follow immediately from Equations 13 and 12, and are given by:

$$\rho_g \partial_t v = -\rho_g v \partial_x v - RT \partial_x \rho_g + \mu \partial_x^2 v - \frac{\mu}{K} v \quad (25a)$$

$$\partial_t \rho_g = -\partial_x (\rho_g v) + \mu_\epsilon \partial_x^2 \rho_g - \dot{m} \quad (25b)$$

$$\partial_t \rho_s = \dot{m} \quad (25c)$$

Where we only added a diffusion term to the mass conservation equation (for stability, but we can keep ϵ small enough to not influence the results), and we applied the ideal gas law, since temperature is assumed to be constant. The initial conditions are given by:

$$v(x, 0) = 0, \quad (26a)$$

$$\rho_g(x, 0) = \rho_{g,0}, \quad (26b)$$

$$\rho_s(x, 0) = \rho_{s,0}. \quad (26c)$$

where we can again find $\rho_{g,0}$ and $\rho_{s,0}$ in the same way as in Model 1. The 0 velocity represents the fact that initially the hydrogen gas is at rest. The boundary conditions for a constant static pressure are given by:

$$v(\partial\Omega_2, t) = 0, \quad (27a)$$

$$\partial_x v(\partial\Omega_1, t) = 0, \quad (27b)$$

$$\rho_g(\partial\Omega_1, t) = \rho_{g,in}, \quad (27c)$$

$$\partial_x \rho_g(\partial\Omega_2, t) = 0. \quad (27d)$$

if we want a constant total pressure, we can change the boundary condition for ρ_g at the inlet to:

$$\rho_g(\partial\Omega_1, t) = \frac{p_{tot} - \frac{1}{2} \rho_g v^2}{RT}. \quad (28)$$

Model 5: 2D Navier-Stokes

The last model will be the extension of our previous model into 2 dimensions. The domain is the same as in Model 3, and the governing equations are given by:

$$\rho_g \partial_t \mathbf{v} = -\rho_g (\mathbf{v} \cdot \nabla) \mathbf{v} - RT \nabla \rho_g + \mu \Delta \mathbf{v} - \frac{\mu}{K} \mathbf{v}, \quad (29a)$$

$$\partial_t \rho_g = -\nabla \cdot (\rho_g \mathbf{v}) + \mu_\epsilon \Delta \rho_g - \dot{m}, \quad (29b)$$

$$\partial_t \rho_s = \dot{m} \quad (29c)$$

which is the 2D version of Equations 25, where the only variable difference is that $\mathbf{v}$ is now a vector. The initial conditions are given by:

$$\mathbf{v}(\mathbf{x}, 0) = 0, \quad (30a)$$

$$\rho_g(\mathbf{x}, 0) = \rho_{g,0}, \quad (30b)$$

$$\rho_s(\mathbf{x}, 0) = \rho_{s,0}. \quad (30c)$$

And the boundary conditions are given by:

$$\mathbf{v}(\partial\Omega_2, t) = 0, \quad (31a)$$

$$\partial_n \mathbf{v}(\partial\Omega_1, t) = 0, \quad (31b)$$

$$\rho_g(\partial\Omega_1, t) = \rho_{g,in}, \quad (31c)$$

$$\partial_n \rho_g(\partial\Omega_2, t) = 0. \quad (31d)$$

where we can also change the boundary condition for ρ_g at the inlet to a total pressure boundary condition, as in Model 4.